

# Introduction

## Introduction

Welcome to the AI course, where we embark on a journey into the technical aspects of Artificial Intelligence. This course is built upon the foundational work of numerous experts in the field, acknowledging that the instructor stands "on the shoulders of giants." The materials presented draw from various sources, with due credit given to the original creators, reflecting a collaborative spirit in the advancement of AI education. This approach allows for the synthesis of diverse perspectives and the integration of compelling insights from across the AI landscape.

Artificial Intelligence is a remarkably broad and multifaceted field, taught in top departments worldwide, yet often with limited overlap in specific topics across institutions. This diversity stems from AI's philosophical roots; it's not a problem with a single or a few solutions, but rather a general philosophical challenge that manifests in countless concrete problems, each with its own array of solutions. Conferences like AAAI (Association for the Advancement of Artificial Intelligence) exemplify this breadth, often featuring numerous parallel tracks, where even seasoned researchers might only grasp a fraction of the presented work. This course aims to provide a comprehensive, yet accessible, introduction to this expansive domain.

## Core Concepts

| Term           | Definition                                                                                                            | Significance                                                                                             |
|----------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Breadth Course | A course designed to introduce a wide range of topics and ideas within a field, rather than focusing deeply on a few. | This course prioritizes exposing students to many AI concepts, preparing them for future specialization. |
| Depth Course   | A course that delves into a few specific topics within a field with significant detail and rigor.                     | Typically pursued after a breadth course, allowing for specialized knowledge and research.               |
| Modeling       | The process of transforming a real-world problem into a simplified, approximated computational problem.               | Crucial first step in solving any real-world problem computationally; often                              |

| Term                         | Definition                                                                                                                                             | Significance                                                                                                 |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
|                              |                                                                                                                                                        | where much "magic" happens.                                                                                  |
| <b>Algorithm</b>             | A step-by-step procedure or formula for solving a computational problem.                                                                               | Provides the method to solve the modeled problem; can be innovative or leverage existing solutions.          |
| <b>Theory</b>                | The mathematical and logical foundations that prove properties, complexities, and guarantees of models and algorithms.                                 | Ensures reliability and understanding of limitations; interacts with both modeling and algorithms.           |
| <b>Application</b>           | The real-world problem or system where AI techniques are deployed and utilized.                                                                        | The ultimate goal of AI development, where theoretical concepts meet practical challenges.                   |
| <b>Philosophical Problem</b> | A fundamental question or challenge that lacks a single, definitive answer and has many interpretations and approaches.                                | Highlights AI's complex nature, extending beyond mere technical solutions to encompass broader implications. |
| <b>Shortest Path Problem</b> | A classic computational problem of finding a path between two nodes in a graph such that the sum of the weights of its constituent edges is minimized. | Used as a concrete example to illustrate the application-modeling-algorithm-theory framework in the lecture. |
| <b>Abstraction</b>           | The process of simplifying complex reality by focusing on relevant details and ignoring irrelevant ones.                                               | Essential in modeling to create manageable computational problems from messy real-world scenarios.           |

## **Detailed Analysis**

### **1. The Course Philosophy: Breadth Over Depth for Foundational Understanding**

This AI course is designed as a breadth course, aiming to introduce students to a wide array of ideas and concepts within Artificial Intelligence. The instructor acknowledges the inherent trade-off between covering many topics broadly and delving deeply into a few. For undergraduate students, the emphasis is placed on gaining a broad understanding, which is considered crucial at this stage of their academic journey. The goal is to equip students with a general point of view and a set of versatile tools that can be applied to various problems, even outside the specific contexts in which they are learned.

The course serves as an introductory gateway, providing foundational knowledge that can be built upon in more specialized studies. For instance, it will cover search algorithms, preparing students for advanced heuristic search courses. It will touch upon basic networks and probabilistic graphical models, laying the groundwork for specialized courses by experts like Daphne Koller. Similarly, an introduction to planning and decision-making will serve as a prerequisite for advanced reinforcement learning courses. The philosophy is that while this course won't make students experts in any single subfield, it will provide the necessary introduction to many, enabling them to pursue deeper studies if they choose to specialize in AI research or industry.

### **2. The Broad and Diverse Nature of AI**

AI is not merely a technical subject; it originates from philosophical inquiries and encompasses a vast range of subfields. The instructor recalls attending AAAI conferences with 8 or 9 parallel tracks, where even as an AI researcher, understanding only one track was common. This illustrates the immense diversity within AI, making it impossible for any single course, especially one with limited time, to do justice to all its facets. The widely used Russell and Norvig textbook, for example, is a "big fat book" that cannot be fully covered in a single class. Therefore, instructors must make deliberate choices about what to include.

The course acknowledges that AI is a general philosophical problem with countless manifestations into concrete problems, each demanding different solutions. This inherent complexity and diversity mean that AI courses across different institutions may have limited overlap in their specific topics. This course embraces this diversity by providing a broad overview, ensuring students are exposed to the fundamental ideas that underpin various AI domains, rather than getting lost in the intricate details of a single niche.

### 3. The Application-Modeling-Algorithm-Theory Framework

A crucial aspect of understanding and solving problems in AI, and indeed in computer science, is to grasp the full spectrum from real-world application to theoretical underpinnings. The instructor introduces a four-step framework: Application, Modeling, Algorithm, and Theory. This framework is illustrated using the example of creating a mapping service like Google Maps.

- **Application (Real-World Problem):** The starting point is a real-world problem, such as finding the best route from location A to location B on a map. This is a practical, tangible challenge faced by users.
- **Modeling (Computational Problem):** The next step is to transform this real-world problem into a computational one. This involves abstraction – simplifying the problem by ignoring irrelevant details (e.g., car type, gas stations, potholes) and focusing on essential elements. For the map example, this means representing the world as a graph where intersections are nodes and road segments are edges. Weights on edges could represent length or average travel time, further abstracting complex factors like traffic, weather, or time of day. This modeling phase is critical, as a well-modeled problem often leads to readily available solutions.
- **Algorithm (Solution to Computational Problem):** Once the problem is modeled computationally (e.g., as a shortest path problem on a graph), existing algorithms can be applied. Examples include Dijkstra's algorithm or Bellman-Ford. The choice of algorithm depends on the specific properties of the modeled problem (e.g., presence of negative edge weights). Sometimes, innovation might be required to develop new algorithms or adapt existing ones.
- **Theory (Proofs and Properties):** The final component involves the theoretical analysis of the model and algorithm. This includes proving properties about the solution (e.g., always finding the shortest path), understanding the complexity class of the problem, and analyzing the algorithm's efficiency and correctness. Theory provides the guarantees and limitations of the chosen approach.

These four components are interconnected: theory informs modeling and algorithms, and algorithms are implemented within the application. The application itself also involves practical considerations like data storage, memory management, and how to handle large-scale data (e.g., not needing the entire world map for a local route).

#### **4. Course Focus: Emphasizing Modeling and Algorithms**

In this particular course, the primary focus will be on the modeling and algorithmic aspects of AI. While applications will be explored, particularly through assignments, and theory will be acknowledged, the deep mathematical proofs and theoretical intricacies will not be the central theme. The instructor will occasionally introduce theorems but will generally not delve into their proofs, leaving that for more advanced, specialized courses.

The emphasis will be on understanding how new problems can be effectively modeled into computational forms, how different models lead to various solutions, and what properties characterize these solutions. This practical orientation aims to equip students with the ability to approach and solve real-world technical problems using an "AI way of thinking," which is valuable whether they pursue a career in AI or in broader industry roles. The course aims to foster a problem-solving mindset, where students learn to break down complex challenges into manageable, computationally solvable components.

#### **Key Takeaways**

- [ ] AI is a broad, diverse field with philosophical roots, not just a technical subject.
- [ ] This course prioritizes breadth, introducing many AI concepts to provide a strong foundation for future specialization.
- [ ] The instructor draws upon the work of many experts, synthesizing diverse materials for the course.
- [ ] Understanding the Application-Modeling-Algorithm-Theory framework is crucial for solving real-world problems.
- [ ] Modeling involves abstracting real-world problems into simplified computational forms, which is often the most critical step.
- [ ] Algorithms provide the computational solutions to modeled problems, often leveraging existing techniques.
- [ ] Theory provides the mathematical guarantees and understanding of an algorithm's properties and limitations.
- [ ] This course will focus primarily on modeling and algorithms, with applications explored in assignments and theory discussed at a high level.
- [ ] The "AI way of thinking" developed in this course is applicable to a wide variety of technical problem-solving scenarios.

## **Conclusion**

This introductory AI course sets the stage for understanding a field that is as philosophically profound as it is technically complex. By emphasizing a breadth of concepts and a practical framework encompassing application, modeling, algorithms, and theory, it equips students with a versatile toolkit for tackling diverse challenges. The focus on modeling and algorithmic thinking, rather than deep theoretical proofs, ensures that undergraduates gain a functional understanding of how AI problems are approached and solved in the real world. This foundational knowledge will serve as a springboard, whether students choose to delve deeper into specific AI subfields through advanced courses or apply these problem-solving methodologies in various industrial and research settings.